

My School
Academic year 2025/2026
1st Secondary Education

Mathematics
2nd Assessment 1st Exam
Proofs

Surname: ..... Name: .....

Group: ..... Date: .....

Question	1	2	Total
Points	5	5	10
Grade			

Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page.

1. Given the equation $x^n + y^n = z^n$ for (x, y, z) and n positive integers.

   (a) (*2 points*) For what values of n is the statement in the previous question true?

   (b) (*3 points*) For $n = 2$ there's a theorem with a special name. What's that name?

   (c) What famous mathematician had an elegant proof for this theorem but there was not enough space in the margin to write it down?

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2. (*5 points*) Prove that the real part of all non-trivial zeros of the function $\zeta(z)$ is $\frac{1}{2}$.

Model A